Junho Andrew Choi

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SUMMARY

Data Science and Integrative Biology senior at UC Berkeley targeting data engineering and data platform roles. Built ingestion pipelines across heterogeneous clinical modalities, mapped and documented a production ERP schema, and audited data quality in a Neo4j graph pipeline. Python, SQL, PostgreSQL, Neo4j, and a habit of measuring whether a pipeline is actually correct.

EDUCATION

University of California, Berkeley Expected May 2027
B.A. Data Science (Concentration: Computational Methods in Biology) and B.A. Integrative Biology *Berkeley, CA*
Coursework: Data Engineering (DS C101, in progress), Machine Learning (CS 189, in progress), Data Structures (CS 61B), Principles & Techniques of Data Science (Data 100), Probability for Data Science (Data C140), Optimization Modeling

SKILLS

Languages: Python (pandas, NumPy), SQL, Java, R
Data Engineering: PostgreSQL, Neo4j, SQLite, schema mapping and documentation, data modeling, ingestion and preprocessing pipelines, data quality auditing, entity resolution and deduplication, JSON/JSONL, retrieval-augmented generation (RAG), natural language to SQL (Text2SQL)
Tools: Git, Jupyter, scikit-learn, PyTorch, React, Vite, Figma
Spoken: Korean (native), English (fluent), Mandarin

EXPERIENCE

SmartMind AI May 2026 - Aug 2026
AI Researcher, Data Science Team *Seoul, Republic of Korea*
- Mapped and documented the production enterprise resource planning (ERP) schema behind a natural-language-to-SQL pipeline, identifying the 6 core tables of 20 required for coverage and capturing the business logic queries depended on (debit/credit balancing, fiscal-period joins, composite keys) by extracting undocumented rules from the client data team
- Audited data quality in a Neo4j-based ontology pipeline (ApeRAG/Qurify) by tuning similarity confidence thresholds and entity-merge accuracy, tracing failed queries back to missing or duplicate graph nodes to guide targeted pipeline fixes
- Replaced single-run pass/fail testing with a 43-run measurement design (15 scenarios, 3 trials each), converting an unverified 20/20 result into a measured 13/15, and corrected 2 errors in the reference answer key by reconciling live query output against source financial statements

Asan Medical Center (AMC) May 2025 - Aug 2025
AI Research Intern *Seoul, Republic of Korea*
- Built automated preprocessing pipelines for 3D CT imaging and clinical records, standardizing ingestion across modalities and cutting manual data preparation time for the research team
- Ran large-scale exploratory analysis across multimodal datasets to identify the data patterns that informed feature selection and a strategy placing 7th among multi-institutional teams

Invention Corps of Berkeley - Mozilla Firefox Aug 2025 - Feb 2026
Design Engineer Consultant *Berkeley, CA*
- Engineered a React and Vite browser extension that ingested live system telemetry to track, filter, and sort 50+ concurrent tabs by memory and CPU usage in real time

PROJECTS

- **Multi-Agent Ad Generation System:** designed a versioned JSON spec as the contract between 8 pipeline stages, backed by a SQLite audience store (500 users, 12 segments), embedding-based catalog retrieval, and deterministic validation gates
- **Geospatial Route Optimizer:** computed pairwise distances across 200+ real-world coordinates and applied depth-first search with branch-and-bound pruning to identify globally optimal traversal paths
- **Heart Disease Predictor and Spam Classifier:** end-to-end pipelines covering preprocessing, feature engineering, and precision-recall tuning (87.3% accuracy on the UCI Cleveland dataset; 8,000+ SpamAssassin examples)

LEADERSHIP & AWARDS

US Army, 8th Army 2ID HHBN Sep 2022 - Mar 2024
Non-Commissioned Officer *Republic of Korea*
- Led a 40+ soldier platoon as a bilingual Korean-English platoon leader, delivering Army Combat Fitness Test (ACFT) training and coursework to a 96% platoon pass rate